EndoTarget Evidence — research decision brief

Decision: hold the candidate pool for further evidence. This public-data benchmark did not produce a statistically supported cross-cohort target shortlist or evidence that network propagation improves on expression ranking. It demonstrates a traceable research decision, including negative results, rather than a validated therapeutic discovery.

Prepared 14 October 2026. Scientific results were executed on SRV01; bounded container reproduction passed locally. A two-page PDF edition accompanies this source.

Question and study design

Does a fixed network-based prioritization improve the transfer of stromal expression associations in endometriosis across public studies? The comparison uses author-adjudicated proliferative-phase eutopic endometrial stromal fibroblasts. Non-endometriosis surgical controls can have other gynecological conditions.

Cohort	Analyzed people	Primary cells / nuclei	Genes tested	BH q < 0.05
Discovery, GSE266265	12 cases + 10 controls	150,788 cells	17,733	0
Replication, HECA E-MTAB-14039	15 cases + 7 controls	52,744 nuclei	17,643	0

Counts were aggregated per donor, then modeled separately by cohort with edgeR normalization and limma-voom, adjusting for age, source batch and adenomyosis. The sample sizes are **22 people per cohort**, not the cell totals. BH q values account for the complete tested gene family; zero significant genes means insufficient evidence under this design, not proof of no disease effect. See the method card.

Frozen comparison and external results

All methods used the same 11,749 discovery-expressed network genes and 123,313 STRING functional-association edges. B0 ranks absolute moderated disease statistics; B1 ranks weighted network degree; G1 propagates discovery evidence through the frozen network. G1 is a restart random walk, with no trained graph neural network or parameter search. Lists, expression directions, parameters and matched reference sets were frozen before calculating HECA disease effects.

Fixed top-20 method	Same direction / evaluable / selected	Same direction among evaluable	Same direction and external q < 0.05
B0: expression	14 / 20 / 20	70.0%	0
B1: network degree	3 / 13 / 20	23.1%	0
G1: propagation	6 / 19 / 20	31.6%	0

Seven B1 genes and one G1 gene failed external expression filtering; none was replaced. B1 misses the prespecified 16/20 evaluability gate. Direction agreement measures estimated expression signs, including small noisy effects; it is not target accuracy or therapeutic success.

G1 minus B0 was **−38.4 percentage points** in the fixed sample. Paired resampling of external donors completed 500 draws: 487 valid and 13 singular-design failures, retained without replacement. The conditional 95% interval was **−38.9 to +8.7 points**, including zero. It covers external sampling sensitivity given the fixed discovery lists/directions, not full discovery-selection or future-experiment uncertainty. No superiority claim follows.

All 22 discovery leave-one-donor-out analyses completed. Median retention of the original 20 genes was B0 14.5, B1 20 and G1 19.5; stable network lists did not yield better external sign agreement. Three viable clinical subset models also yielded no BH-significant genes; two other scenarios failed fixed donor minimums and were not fitted. The 30/50/100-cell thresholds select the same donors, so the primary fits were reused.

Evidence integration and recommendation

All 40 genes in the frozen B0/G1 top-20 union remain visible. SKIDA1 has limited positional/QTL mapping in the prespecified 2023 genetics publication. The other 39 have unavailable mapping in that source, not negative genetic tests. Neither locus assignment nor QTL annotation establishes a causal gene or whether to activate or inhibit it.

The deterministic dossier rule selected only the inconclusive example PARP9: discovery log2 difference +0.466 versus external −0.142, with external 95% interval [−0.413, +0.130] and q = 0.970. Its opposite point-estimate sign does not establish a precise conflict. Recommendation: **hold**, pending better expression precision, cell-state applicability and independent mechanistic evidence. No antibody design, clinical utility or intervention direction is established.

Audit trail and interpretation limits

Five figure topics and linked panels cover donor selection, composition/QC, effect transfer, method sensitivity and the evidence matrix. The full replication report, clinical sensitivities, composition audit and genetic audit provide denominators and provenance.

Discovery reuses author secondary Alevin estimated counts; the full original RDS was not recomputed. HECA has a raw-versus-H5 total discrepancy of 9,607/360,765,431 counts (0.00266%): 1,359 explained and 8,248 not assigned feature by feature. Inherited labels, scRNA/snRNA differences, heterogeneous subtype mixtures and residual clinical confounding limit interpretation. Recruiting sources are separated; genotype-confirmed individual non-overlap is unavailable.

AI assisted implementation and review; methods, evidence and limitations are documented in the audit trail. The local Docker/Nextflow check completed five fresh and five cached synthetic tasks. A real-primary replay reproduced 34 scientific tables and eight JSON result files across five fresh tasks; discrete selections matched exactly and continuous values met absolute tolerance 1e-8. This is engineering reproduction of the same data. The AWS guide remains unexecuted; local scope and receipts document the limits.